

CampusX ML Beginner Notes Lectures 6-10

Lecture 6: Instance-Based vs Model-Based Learning

Explanation: Instance-based memorizes examples; model-based learns a general model.

Key Takeaway: k-NN vs Linear Regression

Practice: Write a short summary and identify one real-world example.

Lecture 7: Challenges in ML

Explanation: Insufficient data, poor quality data, overfitting, underfitting, bias.

Key Takeaway: Clean data is critical.

Practice: Write a short summary and identify one real-world example.

Lecture 8: Applications of ML

Explanation: Healthcare, finance, e-commerce, vision, NLP.

Key Takeaway: Choose ML when patterns exist.

Practice: Write a short summary and identify one real-world example.

Lecture 9: ML Development Life Cycle

Explanation: Problem->Data->Preprocess->Train->Evaluate->Deploy->Monitor.

Key Takeaway: Iterative process.

Practice: Write a short summary and identify one real-world example.

Lecture 10: Framing an ML Problem

Explanation: Define input, output, objective and metric.

Key Takeaway: Regression vs Classification.

Practice: Write a short summary and identify one real-world example.